# Lab 4: Liquid Chemistry

*BIOL-1*

**Name:** _____ **Date:** _____

## Learning Objectives

1. **Define diffusion** and explain how medium density/viscosity affects the rate of diffusion.
2. **Observe viscosity** directly by comparing fluid motion in water versus methyl cellulose.
3. **Demonstrate redox reactions** using a common oxidizer (bleach) and an indicator dye.
4. **Perform serial dilutions** to create a concentration gradient and understand how visual color relates to solute amount.

## Purpose

Perform simplified experiments to observe diffusion, viscosity, redox reactions, and serial dilutions using minimal reagents.

## Connections to Biology

- **Diffusion** is the primary method for moving small molecules (oxygen, carbon dioxide) in and out of cells.
- **Viscosity** of bodily fluids (blood plasma, cytoplasm) affects how easily cells and molecules move through them.
- **Redox reactions** are fundamental to metabolism (cellular respiration and photosynthesis).
- **Concentration gradients** drive many biological processes, including nerve impulses and nutrient absorption.
- **Organic molecules tie-in:** Methyl cellulose is a **carbohydrate polymer** (a cellulose derivative). Its thickness shows how the size and structure of organic macromolecules change the physical behavior of a solution — the same reason a cell's cytoplasm is viscous.

## 1. Diffusion and Viscosity (Water vs. Methyl Cellulose)

**Goal:** Compare properly how red dye diffuses and moves in standard water versus a thick, viscous solution (simulating cytoplasm or mucus).

### Materials

- 2 test tubes
- Distilled water
- Methyl cellulose solution
- Red food color

### Procedure

1. **Fill tube A:** ¾ full with distilled water.
2. **Fill tube B:** Same volume (¾ full) of methyl cellulose solution.
3. Use a straw or dropper to place **one** small drop of red food color at the surface of **Tube A**. **Do not stir.**
4. Place the same size drop at the surface of **Tube B**.
5. Watch for 5–15 minutes. Note how quickly the dye spreads (diffuses/mixes) and if it sinks or stays suspended.

### Questions

**1.** In which tube did the dye spread out faster? Why?

**2.** How would you define **viscosity** based on what you see in Tube B?

**3.** Imagine a cell needs to move oxygen quickly from its membrane to its center. Would this happen faster if the cytoplasm were like water or like methyl cellulose?

## 2. Redox Reaction (Methylene Blue + Bleach)

**Goal:** Show that an oxidizer (bleach) chemically alters and destroys the color of methylene blue.

### Safety

- Work with drops, not large volumes.

### Materials

- 1 test tube
- Methylene blue solution
- 10% bleach
- Distilled water

### Procedure

1. Fill the test tube ½ full with distilled water.
2. Add methylene blue dropwise until the water is a clear, medium blue.
3. Add **1 drop** of bleach and gently swirl.
4. Continue adding bleach one drop at a time, swirling after each addition. Count how many drops it takes until the solution becomes nearly colorless.

### Questions

**1.** What happened to the blue color? Did the dye leave the tube, or did it change form?

**2.** Bleach is an **oxidizing agent**. What does it do to the electrons in the methylene blue molecule?

---

## 3. Potassium Permanganate Serial Dilution

**Goal:** Create a visual concentration gradient to understand how "dilution" works.

### Safety

- Work with drops, not large volumes.

### Materials

- 2 test tubes
- Potassium permanganate stock solution
- Distilled water

### Procedure

1. **Tube 1 (Stock):** Add 5 mL of the **Potassium Permanganate Stock** solution to **Tube 1**.
2. **Tube 2 (Dilution):** Add 5 mL of clear water to **Tube 2**.
3. **Transfer:** Take 1 mL (or a pipette squeeze) from **Tube 1** and add it to **Tube 2**. Mix gently.

### Questions

**1.** Which tube has the highest concentration of potassium permanganate? Which has the lowest?

**2.** Even in the lighter tube, can you still see some pink/purple color? What does this tell you about the molecules in the solution?

**3.** If you took a drop from Tube 2 and put it under a microscope, would you expect to see solid crystals or dissolved ions?